



Mark Meldrum
BY CERIFI

2026

Level 2 - Formula Sheet

This document should be used in conjunction with the corresponding readings in the 2026 Level 2 CFA® Program curriculum. Some of the graphs, charts, tables, examples, and figures are copyright 2025, CFA Institute. Reproduced and republished with permission from CFA Institute. All rights reserved.

Required disclaimer: CFA Institute does not endorse, promote, or warrant accuracy or quality of the products or services offered by MarkMeldrum.com. CFA Institute, CFA®, and Chartered Financial Analyst® are trademarks owned by CFA Institute.

© markmeldrum.com. All rights reserved.



Quantitative Methods

LM1: Basics of Multiple Regression and Underlying Assumptions

LOS: Formulate a MR equation

General Equation: $Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$

k	Slope Coefficients
K + 1	Coefficients
df	Degrees of Freedom = $n - (k + 1) = n - k - 1$
$X_1, X_2 \dots X_n$	Partial Regression Coefficients

Dummy Variables: Take the value of 0 or 1. Use $n - 1$ dummy variables. If all dummy variables = 0, then the regression represents the i^{th} condition (i.e. the intercept).

Log-log Regression: $\ln(Y) = b_0 + b_1\ln(X_1) + b_2\ln(X_2) + \dots + b_n\ln(X_n)$

LM2: Evaluating Regression Model Fit and Interpreting Model Results

LOS: Measures of goodness of fit

R-Squared: $R^2 = \frac{SSR}{SST}$

Adjusted R-Squared: $\bar{R}^2 = 1 - \left(\frac{n-1}{n-k-1} \right) (1 - R^2)$

or: $\bar{R}^2 = 1 - \left[\frac{SSE / (n - k - 1)}{SST / (n - 1)} \right]$

Akaike's information criterion: $n \ln \left(\frac{SSE}{n} \right) + 2(k + 1)$

Schwarz's Bayesian information criterion: $n \ln \left(\frac{SSE}{n} \right) + \ln(n)(k + 1)$

Testing Joint Hypotheses for Coefficients: $F = \frac{(SSE \text{ restricted model} - SSE \text{ unrestricted}) / q}{SSE \text{ unrestricted model} / (n - k - 1)}$



LM3: Model Misspecification

LOS: Heteroskedasticity and Serial Correlation

Breusch-Pagan test for conditional heteroskedasticity:

$$\varepsilon^2 = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

- If no conditional heteroskedasticity, R^2 will be very low
- H_0 : no conditional heteroskedasticity
- $nR^2 \sim \chi^2$, df = number of independent variables, then compare with χ_c^2 (one-tailed test)
- Do not want to reject H_0

Durbin-Watson test for SC:

$$DW \approx 2(1 - r)$$

$$r = \text{corr}(\hat{\varepsilon}_t, \hat{\varepsilon}_{t-1})$$

- H_0 : DW = 2
- $DW \approx 2(1 - r)$ where $r = \text{corr}(\hat{\varepsilon}_t, \hat{\varepsilon}_{t-1})$
- Reject H_0 if $DW < d_L$
- Do not reject H_0 if $DW > d_U$

Breusch-Godfrey test for SC:

Original regression:

$$y_t = b_0 + b_1X_{1t} + b_2X_{2t} + u_t$$

$$\hat{u}_t = a_0 + a_1X_{1t} + a_2X_{2t} + p_1u_{t-1} + e_t$$

- H_0 : No serial correlation
- Chi-square test statistic for lag = 1
- F stat with n-p-k-1 degrees of freedom for p>1

LOS: Explain multicollinearity

Variance Inflation Factor (VIF) test for MC:

$$VIF_j = \frac{1}{1 - R_j^2}$$

- R_j^2 is calculated by regressing each independent variable against the k-1 other independent variables
- $VIF_j > 5$ warrants investigation
- $VIF_j > 10$ indicates serious multicollinearity

LM4: Extensions of Multiple Regression

LOS: Formulate and interpret a logistic regression model

$$\ln \frac{P}{(1-P)} = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n + \varepsilon$$

Logistic Regression:

$$P = \frac{1}{1 + \exp\left[-(b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n)\right]}$$



LM5: Time Series Analysis

LOS: Calculate and evaluate trend models

Linear: $y_t = b_0 + b_1t + \varepsilon_t$

If SC, DW $\neq$ 2, SE biased downwards

Log-linear: $\ln(y_t) = b_0 + b_1t + \varepsilon_t$

b_1	Trend Coefficient
t	Time
ε_t	Error Term

LOS: Describe the structure of AR(p)

AR(1) $x_t = b_0 + b_1x_{t-1} + \varepsilon_t$

AR(2) $x_t = b_0 + b_1x_{t-1} + b_2x_{t-2} + \varepsilon_t$

AR(p) $x_t = b_0 + b_1x_{t-1} + b_2x_{t-2} + \dots + b_px_{t-p} + \varepsilon_t$

Covariance Stationary

- Constant mean, variance and Covariance(x_t, x_{t-s}) in all periods. If $b_1 < 1$, then the series is mean reverting and is Covariance stationary. All Cov. St. T have a finite mean reverting level. If not, then move on to first-differencing.
- Cannot use the DW statistic anymore to test for SC (once the model has as IV that is a lagged value of the DV). Use the t-statistics of the autocorrelations of the error terms.

LOS: Calculate mean-reverting level

$$x_t = \frac{b_0}{(1 - b_1)}$$

LOS: Calculate the root mean squared error

The square root of the average squared error:

$$RMSE = \sqrt{\frac{\sum \varepsilon^2}{n}}$$

The smaller the RMSE, the more accurate the model.

LOS: Unit roots, random walks

Random Walk: $b_0 = 0$ and $b_1 = 1$

After first differencing:

$b_0 = 0$ and $b_1 = 0$

(Cov. St., but no forecasting power)

- Cannot estimate an AR(1) on a random walk
- Test for a RW: Dickey-Fuller
- $H_0: g = 0$, $H_a: g < 0$, where $g = (b_1 - 1)$

Random Walk with drift: $b_0 \neq 0$

LOS: ARCH models

$$\hat{\varepsilon}_t^2 = a_0 + a_1 \hat{\varepsilon}_{t-1}^2 + \mu$$

Test: $H_0: a_1 = 0$ versus $H_a: a_1 \neq 0$

Rejecting H_0 implies ARCH(1)

Thus, if $a_1 = 0$, we have constant variance

LOS: Two time-series

Engle-Granger DF test: $(\varepsilon_t - \varepsilon_{t-1}) = b_0 + g\varepsilon_{t-1} + \mu$

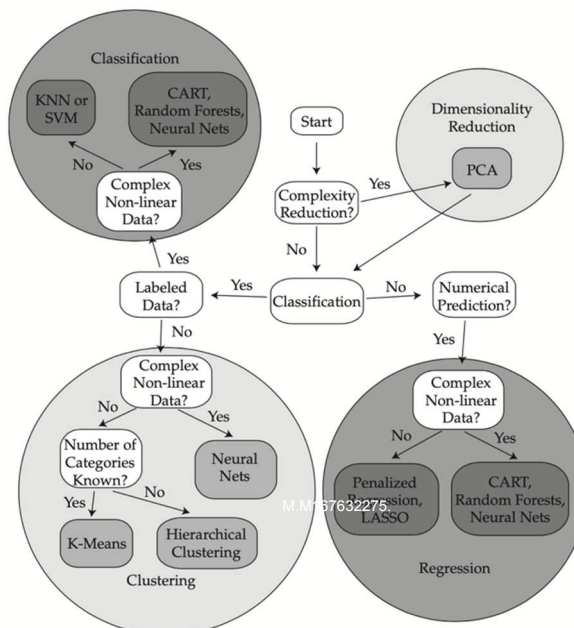
- $H_0: g = 0$, $H_a: g = 1$
- Reject H_0 , regression output is OK.

DV and IV are both TS

Neither TS has a unit root	No	All else, not OK
Both have a unit root but are cointegrated	issues	

LM6: Machine Learning

LOS: supervised versus unsupervised





LM7: Big Data Projects

LOS: Evaluate fit of a ML algorithm

Precision (P) = $TP / (TP + FP)$

Recall (R) = $TP / (TP + FN)$

Accuracy = $(TP + TN) / (TP + FP + TN + FN)$

F1 score = $(2 \times P \times R) / (P + R)$

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(Predicted_i - Actual_i)^2}{n}}$$

where:

FP = false positive

FN = false negative

TP = true positive

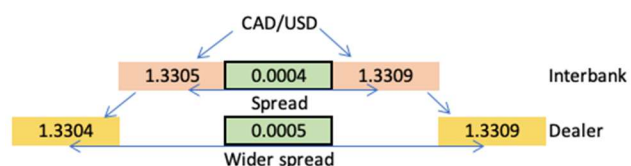
TN = true negative

Economics

LM1: Currency Exchange Rates: Understanding Equilibrium Value

LOS: Calculate and interpret bid-offer spread

P/B = Price/Base



The Dealer will buy 1 USD for 1.3304 CAD & sell 1 USD for 1.3309 CAD

- Dealer bid < Interbank Offer
- Dealer Offer > Interbank bid

Inverting

P/B	bid	offer
B/P	1/offer	1/bid

LOS: Calculate forward premium/discount

Covered interest rate parity (CIRP):

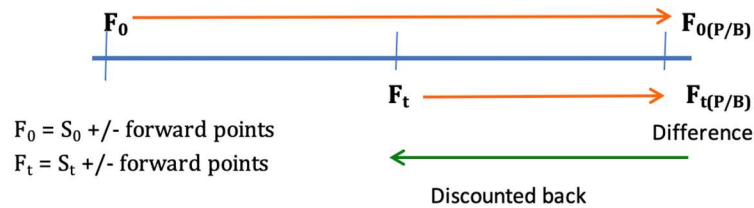
$$F_{P/B} = S_{P/B} \left(\frac{1 + i_P \left[\frac{Actual}{360} \right]}{1 + i_B \left[\frac{Actual}{360} \right]} \right)$$

Premium/discount = $F_{P/B} - S_{P/B}$

If: $i_P > i_B$ then premium, else discount

CIRP holds due to the existence of arbitrage opportunities if it did not.

LOS: Calculate mark-to-market value of a forward



- Step 1: Calculate F_0
- Step 2: Calculate F_t
- Step 3: Discount the difference back to time t

LOS: UIRP, PPP and the International Fisher effect

UIRP - uncovered interest rate parity

$$\% \Delta S_{f/d}^e = i_f - i_d$$

- The expected change in the spot exchange rate over the investment horizon should be reflected in the interest rate differential.

Forward rate parity:

$$F_{f/d} = S_{f/d}^e$$

- The forward exchange rate will be an unbiased forecast of the future spot exchange rate if both CIRP and UIRP hold.

Purchasing Power Parity: (relative)

$$\% \Delta S_{f/d} \cong \pi_f - \pi_d$$

- The percentage change in the spot exchange rate will be completely determined by the difference between the foreign and domestic inflation rates.

Purchasing Power Parity: (ex ante)

$$\% \Delta S_{f/d}^e = \pi_f^e - \pi_d^e$$

- The expected percentage change in the spot exchange rate will be completely determined by the difference between the foreign and domestic expected inflation rates.

International Fisher Effect:

$$i_f - i_d = \pi_f^e - \pi_d^e$$

- Combines both URIP and PPP(ex ante)
- If real interest rates are equal across markets, the foreign-domestic nominal yield spread is determined solely by the foreign-domestic expected inflation differential.

LM2: Economic Growth

LOS: Relation of LR stock market appreciation & economic growth rate

$$(1/T)\% \Delta P = (1/T)\% \Delta \text{GDP} + (1/T)\% \Delta (E/\text{GDP}) + (1/T)\% \Delta (P/E)$$

P	aggregate value of equities
E/GDP	corporate earnings as a share of GDP
P/E	price-to-earnings ratio of the market



- Over short to immediate horizons, all three of these factors contribute to appreciation or depreciation of the stock market. In the long run, however, the growth rate of GDP must dominate. The second term in the equation above must be approximately zero over long horizons. The third term must also be approximately zero over long horizons. The conclusion is that the drivers of potential GDP are ultimately the drivers of stock market price performance.

LOS: Contrast capital deepening versus technological progress

$$Y = AK^{\alpha}L^{1-\alpha} \quad \text{and} \quad y = Ak^{\alpha}$$

Y	Aggregate Output
A	Total Factor Productivity
K	Capital Stock
L	Quantity of Labour
α	Share of Output paid to Capital

- An increase in k = capital deepening
- An increase in A = technological Progress
- The amount of goods a worker can produce (labour productivity) depends on the amount of capital available for each worker (capital-to-labour ratio), technology or TFP, and the share of capital in GDP (α).

LOS: Growth of accounting relations

Growth accounting equation: $\frac{\Delta Y}{Y} \approx \frac{\Delta A}{A} + \alpha \frac{\Delta K}{K} + (1 - \alpha) \frac{\Delta L}{L}$

- The growth rate of output equals the rate of technological change plus α times the growth rate of capital plus $(1 - \alpha)$ times the growth rate of labour.

Labour productivity growth accounting equation:

Growth rate in potential GDP =

Long-term growth rate of labour force + Long-term growth rate in labour productivity

- Thus, potential GDP growth is a combination of the long-term growth rate of the labour force and the long-term growth rate of labour productivity. If the labour force is growing at 1 percent per year and productivity per worker is rising at 2 percent per year, then potential GDP is rising at 3 percent per year.

Balanced or steady state rate of growth

$$\text{Growth rate of output per capita} = \theta / (1 - \alpha)$$

$$\text{Growth rate of output} = \theta / (1 - \alpha) + n$$



Financial and Reporting Analysis

Candidates should be aware that certain ratios may be defined differently. Such differences are part of the nature of practical financial analysis. For examination purposes, when alternative ratio definitions exist and no specific definition is given in the question, candidates should use the definition provided in this list of ratios.

1. Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$
2. Quick Ratio = $\frac{[\text{Cash} + \text{Short-term marketable investments} + \text{Receivables}]}{\text{Current Liabilities}}$
3. Cash Ratio = $\frac{[\text{Cash} + \text{Short-term marketable investments}]}{\text{Current Liabilities}}$
4. Defensive Interval Ratio = $\frac{[\text{Cash} + \text{Short-term marketable investments} + \text{Receivables}]}{\text{Daily Cash Expenditures}}$
5. Receivables Turnover Ratio = $\frac{\text{Total Revenue}}{\text{Average Receivables}}$
6. Days of Sales Outstanding (DSO) = $\frac{[\# \text{ of days in period}]}{\text{Receivables Turnover Ratio}}$
7. Inventory Turnover Ratio = $\frac{\text{COGS}}{\text{Average Inventory}}$
8. Days of Inventory on hand (DOH) = $\frac{\# \text{ of Days in Period}}{\text{Inventory Turnover Ratio}}$
9. Payables Turnover Ratio = $\frac{\text{Purchases}}{\text{Average Trade Payables}}$
10. Number of Days of Payables = $\frac{\text{Number of Days in Period}}{\text{Payables Turnover Ratio}}$
11. Cash Conversion Cycle (Net Operating Cycle) = DOH + DSO – Number of Days of Payables
12. Working Capital Turnover Ratio = $\frac{\text{Total revenue}}{\text{Average working capital}}$
13. Fixed Asset Turnover Ratio = $\frac{\text{Total Revenue}}{\text{Average Net Fixed Assets}}$
14. Total Asset Turnover Ratio = $\frac{\text{Total revenue}}{\text{Average total assets}}$
15. Gross Profit Margin = $\frac{\text{Gross Profit}}{\text{Total revenue}}$
16. Operating Profit Margin = $\frac{\text{Operating Profit}}{\text{Total revenue}}$
17. Pre-Tax Margin = $\frac{\text{Earnings before tax but after interest}}{\text{Total Revenue}}$
18. Net Profit Margin = $\frac{\text{Net Income}}{\text{Total Revenue}}$
19. Operating Return on Assets = $\frac{\text{Operating Income}}{\text{Average Total Assets}}$



20. Return on Assets = $\frac{\text{Net Income}}{\text{Average Total Assets}}$
21. Return on Equity = $\frac{\text{Net Income}}{\text{Average Shareholders' Equity}}$
22. Return on Total Capital = $\frac{\text{Earnings before Interest and Taxes}}{[\text{Interest bearing Debt} + \text{Shareholders' Equity}]}$
23. Return on Common Equity = $\frac{[\text{Net income} - \text{Preferred dividends}]}{\text{Average Common Shareholders' Equity}}$
24. Tax Burden = $\frac{\text{Net Income}}{\text{Earnings before Taxes}}$
25. Interest Burden = $\frac{\text{Earnings before Taxes}}{\text{Earnings before Interest and Taxes}}$
26. EBIT Margin = $\frac{\text{Earnings before Interest and Taxes}}{\text{Total Revenue}}$
27. Financial Leverage Ratio (Equity Multiplier) = $\frac{\text{Average Total Assets}}{\text{Average Shareholders' Equity}}$
28. Total debt = The total of interest-bearing short-term and long-term debt, excluding liabilities such as accrued expenses and accounts payable
29. Debt-to-Asset Ratio = $\frac{\text{Total Debt}}{\text{Total Assets}}$
30. Debt-to-Equity Ratio = $\frac{\text{Total Debt}}{\text{Total Shareholders' Equity}}$
31. Debt-to-Capital Ratio = $\frac{\text{Total Debt}}{[\text{Total Debt} + \text{Total Shareholders' Equity}]}$
32. Interest Coverage Ratio = $\frac{\text{Earnings before Interest and Taxes}}{\text{Interest Payments}}$
33. Fixed Charge Coverage Ratio = $\frac{[\text{Earnings before Interest and Taxes} + \text{Lease Payments}]}{[\text{Interest payments} + \text{Lease payments}]}$
34. Dividend Payout Ratio = $\frac{\text{Common Share Dividends}}{\text{Net Income attributable to Common Shares}}$
35. Retention Rate = $\frac{[\text{Net income attributable to common shares} - \text{Common share dividends}]}{\text{Net income attributable to common shares}} = 1 - \text{Payout Ratio}$
36. Sustainable growth rate = Retention rate $\times$ Return on Equity
37. Earnings per Share = $\frac{[\text{Net income} - \text{Preferred dividends}]}{\text{Weighted Average number of Ordinary Shares Outstanding}}$
38. Book value per Share = $\frac{\text{Common Stockholders' Equity}}{\text{Total Number of Common Shares Outstanding}}$
39. Free cash flow to equity (FCFE) = Cash flow from operating activities – Investment in fixed capital + Net borrowing



40. Free cash flow to the firm (FCFF) = Cash flow from operating activities + Interest expense $\times$ (1 – Tax rate) – Investment in fixed capital (Interest expense should be added back only if it was subtracted in determining cash flow from operating activities. This may not be the case for companies electing an alternative treatment under IFRS.)

LM1: Intercorporate Investments

Equity Method:

Investment in Associates	→ On the BS
+ %'age of Net Income	→ Reported on IS
- %'age of Dividends	→ Not reported as income

Purchase Price

- %'age of BV
= Excess Purchase Price
- %'age of FV attributable to assets
= Goodwill → *Tested annually for impairment*

Investment in Associates
+ %'age of Net Income
- %'age of Dividends
- amortization of %'age of FV attributable to assets
- %'age unrealized profit (current period upstream sales)
+ previous unrealized profits now realized

Acquisition Method:

Partial Goodwill (IFRS only)

FV of consideration → Price paid
- %'age of FV of net identifiable assets
= Goodwill recognized

Full Goodwill (GAAP and IFRS)

FV of entity: (1/%'age purchased) $\times$ Purchase Price
- FV of net identifiable assets
= Goodwill recognized

Non-Controlling Interest: Full Goodwill

%'age minority interest $\times$ FV of the firm

Non-Controlling Interest: Partial Goodwill

%'age minority interest $\times$ FV of net identifiable assets

Goodwill Impairment: IFRS

Recoverable amount of unit
- Carrying value of unit
= Impairment loss



Goodwill Impairment: GAAP

FV of reporting unit

- net assets

= Implied Goodwill

(Current carrying value of goodwill – Implied Goodwill =
Impairment Loss)

LM2: Employee Compensation: Post-Employment and Share-Based

LOS: Explain and calculate

Funded status = Fair Value of the plan assets - PV of the Defined Benefit Obligation

IFRS Component	IFRS Recognition	US GAAP Component	US GAAP Recognition
Service costs	Recognised in P&L.	Current service costs	Recognised in P&L.
		Past service costs	Recognised in OCI and subsequently amortised to P&L over the service life of employees.
Net interest income/expense	Recognised in P&L as the following amount: Net pension liability or asset $\times$ interest rate ^a	Interest expense on pension obligation	Recognised in P&L.
		Expected return on plan assets	Recognised in P&L as the following amount: Plan assets $\times$ expected return.
Remeasurements: Net return on plan assets and actuarial gains and losses	Recognised in OCI and not subsequently amortised to P&L. - Net return on plan assets = Actual return – (Plan assets $\times$ Interest rate). - Actuarial gains and losses = Changes in a company's pension obligation arising from changes in actuarial assumptions.	Actuarial gains and losses including differences between the actual and expected returns on plan assets	Recognised immediately in P&L or, more commonly, recognised in OCI and subsequently amortised to P&L using the corridor or faster recognition method. ^b - Difference between expected and actual return on assets = Actual return – (Plan assets $\times$ Expected return). - Actuarial gains and losses = Changes in a company's pension obligation arising from changes in actuarial assumptions.

Periodic Pension Cost = Ending Funded Status - Employer Contributions - Beginning Funded Status

Actuarial Assumptions:

- Estimated final salary
- Life Span in Retirement
- Discount rate

LM3: Multinational Operations

LOS: Calculate translation effects

Net income in FC	[Translated according to the method used to translate the income statement]	=	Net income in PC
– Dividends in FC	$\times$ Exchange rate when dividends declared	=	– Dividends in PC
<u>R/E in FC</u>			<u>R/E in PC</u>
Beginning R/E in FC	[From last year's translation]	$\rightarrow$	Beginning R/E in PC
+ Net income in FC	[Translated according to method used to translate the income statement]	=	+ Net income in PC
– Dividends in FC	$\times$ Exchange rate when dividends declared	=	– Dividends in PC
<u>Ending R/E in FC</u>			<u>Ending R/E in PC</u>



Translation method:	Foreign Subsidiary's Functional Currency	
	Foreign Currency	Parent's Presentation Currency
	Current Rate Method	Temporal Method
Exchange rate at which financial statement items are translated from the foreign subsidiary's bookkeeping currency to the parent's presentation currency:		
Assets		
Monetary, such as cash and receivables	Current rate	Current rate
Non-monetary		
▪ measured at current value (e.g., marketable securities and inventory measured at market value under the lower of cost or market rule)	Current rate	Current rate
▪ measured at historical costs, (e.g., inventory measured at cost under the lower of cost or market rule; property, plant & equipment; and intangible assets)	Current rate	Historical rates
Liabilities		
Monetary, such as accounts payable, accrued expenses, long-term debt, and deferred income taxes	Current rate	Current rate
Non-monetary		
▪ measured at current value	Current rate	Current rate
▪ not measured at current value, such as deferred revenue	Current rate	Historical rates
Equity		
Other than retained earnings	Historical rates	Historical rates
Retained earnings	Beginning balance plus translated net income less dividends translated at historical rate	Beginning balance plus translated net income less dividends translated at historical rate
Revenues		
Average rate	Average rate	Average rate
Expenses		
Most expenses	Average rate	Average rate
Expenses related to assets translated at historical exchange rate, such as cost of goods sold, depreciation, and amortization	Average rate	Historical rates
Treatment of the translation adjustment in the parent's consolidated financial statements	Accumulated as a separate component of equity	Included as gain or loss in net income

	Temporal Method, Net Monetary Liability Exposure	Temporal Method, Net Monetary Asset Exposure	Current Rate Method
Foreign currency strengthens relative to parent's presentation currency	↑ Revenues ↑ Assets ↑ Liabilities ↓ Net income ↓ Shareholders' equity Translation loss	↑ Revenues ↑ Assets ↑ Liabilities ↑ Net income ↑ Shareholders' equity Translation gain	↑ Revenues ↑ Assets ↑ Liabilities ↑ Net income ↑ Shareholders' equity Positive translation adjustment
Foreign currency weakens relative to parent's presentation currency	↓ Revenues ↓ Assets ↓ Liabilities ↑ Net income ↑ Shareholders' equity Translation gain	↓ Revenues ↓ Assets ↓ Liabilities ↓ Net income ↓ Shareholders' equity Translation loss	↓ Revenues ↓ Assets ↓ Liabilities ↓ Net income ↓ Shareholders' equity Negative translation adjustment

- Typically, under the current rate method, Total assets > Total liabilities, therefore the subsidiary will have net asset balance sheet exposure. The foreign currency strengthened throughout the year (\$1 CAD bought 0.7 EUR on Jan 1 but 0.80 on Dec 31) thus we have a positive translation adjustment on the balance sheet. If you reverse the exchange rates in the table above, you will see that we arrive at a negative translation adjustment (since we are simulating a weakening CAD).



- Under the temporal method, if total exposed assets > total exposed liabilities, the subsidiary will have net asset balance sheet exposure (net liability balance sheet exposure otherwise). The foreign currency strengthened throughout the year (\$1 CAD bought 0.7 EUR on Jan 1 but 0.80 on Dec 31) and we have a translation gain on the income statement. This implies net asset balance sheet exposure. If you reverse the exchange rates in the table above, you will see that we arrive at a translation loss on the IS (since we are simulating a weakening CAD).

LM4: Analysis of Financial Institutions

LOS: Explain CAMELS

Capital Adequacy ratios

Common Equity Tier 1 Capital/risk-weighted Assets	[at least 4.5%]
Tier 1 Capital/risk-weighted Assets	[at least 6%]
Total Capital/risk-weighted Assets	[at least 8%]

LOS: Key ratios for analyzing insurance companies

Loss and loss adjustment expense ratio = (Loss expense + Loss adjustment ratio)/Net Premiums Earned

Underwriting expense ratio = Underwriting expense/Net Premiums written

Combined ratio = Loss and loss adjustment expense ratio + Underwriting expense ratio

Dividends to Policyholders (shareholders) ratio = Dividends to policyholders (shareholders)/Net Premiums earned

Combined ratio after dividends = Combined ratio + Dividends to policyholders (shareholders) ratio

LM5: Evaluating Quality of Financial Reports

LOS: Evaluate quality of financial reports

Beneish Model A 'probability of earnings manipulation' model

$$\begin{aligned} \text{M-score} = & -4.84 + 0.920(\text{DSR}) \\ & + 0.528(\text{GMI}) \\ & + 0.404(\text{AQI}) \\ & + 0.892(\text{SGI}) \\ & + 0.115(\text{DEPI}) \\ & - 0.172(\text{SGAI}) \\ & + 4.679(\text{Accruals}) \\ & - 0.327(\text{LEVI}) \end{aligned}$$

DSR (days sales receivable index) $(\text{Receivables}_t / \text{Sales}_t) / (\text{Receivables}_{t-1} / \text{Sales}_{t-1})$

GMI (gross margin index) $\text{Gross margin}_{t-1} / \text{Gross margin}_t$

AQI (asset quality index) $1 - (\text{PPE}_t + \text{CA}_t) / \text{TA}_t / [1 - (\text{PPE}_{t-1} + \text{CA}_{t-1}) / \text{TA}_{t-1}]$



SGI (sales growth index)	$Sales_t / Sales_{t-1}$
DEP (depreciation index)	$Depreciation\ rate_{t-1} / Depreciation\ rate_t$
SGAI (sales, general and administrative index)	$(SGA_t / Sales_t) / (SGA_{t-1} / Sales_{t-1})$
Accruals	$(Income\ before\ extraordinary\ items - Cash\ from\ operations) / Total\ Assets$
LEVI (leverage index)	$Leverage_t / Leverage_{t-1}$

LOS: Describe sustainable earnings

$$Earnings_{t+1} = a + \beta_1 Earnings_t + \varepsilon$$

- A higher coefficient (β_1) represents more persistent earnings

$$Earnings_{t+1} = a + \beta_1 Cash\ Flow_t + \beta_2 Accruals_t + \varepsilon$$

- If $\beta_1 > \beta_2$, cash-flow component of earnings is more persistent

LOS: Evaluate earnings quality

Altman Model A bankruptcy prediction model

$$\begin{aligned} Z\text{-score} = & 1.2(\text{Net working capital} / \text{Total assets}) \\ & + 1.4(\text{Retained earnings} / \text{Total assets}) \\ & + 3.3(\text{EBIT} / \text{Total assets}) \\ & + 0.6(\text{Market value of Equity} / \text{Book value of liabilities}) \\ & + 1.0(\text{Sales} / \text{Total assets}) \end{aligned}$$

The ratios in the model reflect liquidity, profitability, leverage and activity. The higher the Z-score, the better.

Corporate Issuers

LM1: Analysis of Dividends and Share Repurchases

LOS: Calculate effective tax rates

Effective Tax rate for Double Taxation Systems

$$EffTxRate_{DblTx} = 1 - [EBT(1 - T_c)(1 - T_d)] / EBT$$

Effective Tax rate for Split-Rate Taxation Systems

$$EffTxRate_{SptTx} = 1 - [EBT(1 - T_{CL})(1 - T_d)] / EBT$$

$EffTxRate_{DblTx}$	Effective tax rate under double taxation systems
$EffTxRate_{SptTx}$	Effective tax rate under split-rate taxation system
EBT	Earnings before tax (corporate level)
T_c	Corporate statutory tax rate on retained earnings
T_d	Investor tax rate on dividends received
T_{CL}	Corporate statutory tax rate on dividends



LOS: Calculate dividends under different payout policies

Target Payout Adjustment Model

Expected increase in dividends = (Expected earnings × Target Payout Ratio – Previous dividend) × Adjustment Factor

LOS: Calculate effect of repurchase on EPS

EPS after buyback = (Earnings - After-tax cost of funds)/Shares outstanding after buyback

LOS: Calculate effect of repurchase on BVPS

$BVPS_{\text{after}} = (\text{Equity} - \text{buyback amount}) / (\text{PreBuybackShareCount} - \# \text{ of shares repurchased})$

Dividend Coverage ratio = Net Income/Dividends

FCFE Coverage ratio = FCFE/[Dividends + Share repurchases]

LM3: Cost of Capital: Advanced Topics

LOS: Forward-looking approaches to estimating equity risk premium

Grinold-Kroner Model

$ERP = [DY + \text{Expected repricing} + \text{Earnings growth per share}] - E(r_f)$

Constant Growth Dividend Discount Model

$ERP = E\left(\frac{D_1}{V_0}\right) + E(g) - r_f$

LOS: Methods for estimating the required return on equity

Bond Yield Plus Risk Premium Approach

$r_e = r_d + RP$

CAPM

$r_e = r_f + \hat{\beta}(ERP)$

Fama-French Five Factor Model

$r_e = r_f + \beta_1 ERP + \beta_2 SMB + \beta_3 HML + \beta_4 RMW + \beta_5 CMA$



Equity

LM2: Discounted Dividend Valuation

LOS: Calculate/interpret DDM

Dividend Discount Model

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t} = \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \frac{P_t}{(1+r)^n}$$

LOS: Calculate GGM and implied growth rate

Gordon Growth Model

$$V_0 = \frac{D_1}{r-g} = \frac{D_0(1+g)}{r-g} \quad \& \quad r = \frac{D_1}{P_0} + g \quad \text{Total Return} = \text{Dividend Yield} + \text{Capital Gains}$$

- Assumes dividends grow indefinitely at a constant rate
- r must be strictly greater than g
- Both r and g should reflect long-term expectations
- If $g > \text{GDP growth}$, then use a multi-stage DDM

LOS: Calculate/Interpret PVGO

Present Value of a Growth Opportunity:

$$V_0 = P_0 = \frac{E_1}{r} + PVGO$$

Leading P/E

$$\frac{P}{E_1} = \frac{1}{r} + \frac{PVGO}{E_1}$$

LOS: Calculate/Interpret justified P/E

Justified trailing P/E

$$\frac{P_0}{E_0} = \frac{D_0(1+g)/E_0}{r-g} = \frac{(1-b)(1+g)}{r-g}$$

Justified forward P/E

$$\frac{P_0}{E_1} = \frac{1-b}{r-g}$$

LOS: Calculate perpetual preferred (perpetuity)

Perpetuity

$$V_0 = \frac{D}{r}$$



LOS: Calculate/interpret H-model, Two-stage

Two-stage DDM

$$V_0 = \sum_{t=1}^n \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{D_0(1+g_s)^n(1+g_L)}{(1+r)^n(r-g_L)}$$

H-Model

$$V_0 = \frac{D_0(1+g_L) + D_0H(g_s - g_L)}{r - g_L}$$

Three Stage DDM

$$V_0 = \sum_{a=1}^m \frac{D_0(1+g_s)^a}{(1+r)^a} + \sum_{b=m+1}^n \frac{D_0(1+g_s)^m(1+g_t)^b}{(1+r)^{m+b}} + \frac{D_0(1+g_s)^m(1+g_t)^b(1+g_L)}{(1+r)^n(r-g_L)}$$

D	Dividends
P	Price
V	Intrinsic Value
g	Growth Rate
g_s	Short-term growth rate (high growth period)
g_t	Mid-term growth rate (transition growth period)
g_L	Long-term growth rate (low growth period)
b	(1 - dividend payout ratio) = Retention ratio
H	Half-life in years of the high-growth period
RR	Retention Ratio
r	Required Rate of Return

LOS: Calculate/Interpret sustainable growth rate

$$g = RR \times ROE = \frac{\text{Net Income} - \text{Divid}}{\text{Net Income}} \times ROE$$

$$ROE = \frac{\text{Net Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total Assets}} \times \frac{\text{Total Assets}}{\text{Shareholder's Equity}}$$

$$g = \frac{\text{Net Income} - \text{Dividend}}{\text{Net Income}} \times \frac{\text{Net Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total Assets}} \times \frac{\text{Total Assets}}{\text{Shareholder's Equity}}$$

LM3: Free Cash Flow Valuation

LOS: Compare FCFF and FCFE

$$\text{Firm Value} = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1+wa)^t}$$

Equity value = Firm value – Market value of debt

$$\text{Equity Value} = \sum_{t=1}^{\infty} \frac{FCFE_t}{(1+r)^t}$$



LOS: Explain and calculate FCFF/FCFE from NI/EBIT/EBITDA/CFO

$$FCFF = NI + NCC + \text{Int}(1 - \text{Tax Rate}) - \text{FCInv} - \text{WCInv}$$

$$FCFF = \text{CFO} + \text{Int}(1 - \text{Tax Rate}) - \text{FCInv}$$

$$FCFF = \text{EBIT}(1 - \text{Tax Rate}) + \text{Dep} - \text{FCInv} - \text{WCInv}$$

$$FCFF = \text{EBITDA}(1 - \text{Tax Rate}) + \text{Dep}(\text{Tax Rate}) - \text{FCInv} - \text{WCInv}$$

$$FCFE = NI + NCC - \text{FCInv} - \text{WCInv} + \text{Net Borrowing}$$

$$FCFE = \text{CFO} - \text{FCInv} + \text{Net Borrowing}$$

$$FCFE = \text{EBIT}(1 - \text{Tax Rate}) + \text{Dep} - \text{FCInv} - \text{WCInv} - \text{Int}(1 - \text{Tax Rate}) + \text{Net Borrowing}$$

$$FCFE = \text{EBITDA}(1 - \text{Tax Rate}) + \text{Dep}(\text{Tax Rate}) - \text{FCInv} - \text{WCInv} - \text{Int}(1 - \text{Tax Rate}) + \text{Net Borrowing}$$

$$FCFE = NI - (1 - \text{DR})(\text{FCInv} - \text{Dep}) - (1 - \text{DR})\text{WCInv}$$

- FCFE equals NI minus the amount of fixed capital expenditure (net of depreciation) and working capital investment that is financed by equity. We have assumed that the only noncash charge is depreciation.

LOS: Explain and estimate using single-stage/2-stage/3-stage models

Single-stage (stable growth):

$$\text{Firm Value} = \frac{FCFF_1}{wacc - g} = \frac{FCFF_0(1+g)}{wacc - g}$$

$$\text{Equity Value} = \frac{FCFE_1}{r - g} = \frac{FCFE_0(1 + g)}{r - g}$$

- Note that g in this model need not be the same as the g you may use in a DDM. However, If dividends = FCFE, and g is the same in both models, both will yield the same estimate of value

Two-stage:

$$\text{Firm Value} = \sum_{t=1}^n \frac{FCFF_t}{(1 + wacc)^t} + \frac{FCFF_{n+1}}{(wacc - g)} \frac{1}{(1 + wacc)^n}$$

$$\text{Equity Value} = \sum_{t=1}^n \frac{FCFE_t}{(1+r)^t} + \frac{FCFE_{n+1}}{(r-g)} \frac{1}{(1+r)^n}$$

Value of firm = Value of operating assets + Value of non-operating assets



LM4: Market-Based Valuation: Price and Enterprise Value Multiples

LOS: Calculate/Interpret justified price multiple

Justified forward PE:

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r-g} = \frac{DPR}{r-g}$$

Justified trailing PE:

$$\frac{P_0}{E_0} = \frac{D_0(1+g)/E_0}{r-g} = \frac{DPR(1+g)}{r-g}$$

LOS: Calculate/Interpret price multiples

Price/Earnings	P/E	E/P	Earnings Yield
Price-to-book	P/B		
Price-to-sales	P/S		
Price-to-cashflow	P/CF	CF/P	Cash Flow Yield
Price-to-Dividends	P/D	D/P	Dividend Yield

- Trailing P/E - most recent 4 quarters
- Forward P/E - next year's expected earnings
- Diluted EPS compensates for differences in convertible securities
- Core EPS - adjusts Diluted EPS for non-recurring items

LOS: Calculate/Interpret normalized EPS

Normalized EPS (historical average EPS) $\frac{\sum_{i=1}^n EPS_i}{n}$

- Does not account for changing firm size

Normalized EPS (average ROE) $\frac{\sum_{i=1}^n ROE_i}{n} \times BVPS_n$

- Only if book value makes sense

LOS: Calculate/Interpret justified multiples

Justified P/B $\frac{P_0}{B_0} = \frac{ROE(1-b)}{r-g} = \frac{ROE-g}{r-g}$

Justified P/S $\frac{P_0}{S_0} = \frac{E_0/S_0(1-RR)(1+g)}{r-g}$

LOS: Calculate/Interpret PEG ratio

Forward PEG = $(P_0/E_1)/g$

Trailing PEG = $(P_0/E_0)/g$

- Assumes a linear relationship between PE and g
- Ignores risk



- Does not account for differences in the duration of g

LOS: Calculate/Interpret price multiples in determining terminal values

$$V_n = \text{benchmark } P/E_{ttm} \times E_n$$

$$V_n = \text{benchmark } P/E_{ntm} \times E_{n+1}$$

- Terminal price based on comparables

LOS: Calculate/Interpret EV multiples

EV/EBITDA

where:

$$EV = MV_e + MV_d + MV_{pr} - \text{Cash/Equiv.} - \text{Marketable Securities}$$

Possible variants:

EV/FCFF

EV/EBITDAR

EV/EBITA

EV/EBIT

EV/Sales

R = rent expense

where amortization is the major NCC

DA is not a major item

Controls for differing capital structures that P/S would not

LOS: Describe momentum indicators

Unexpected earnings

$$UE_t = EPS_t - E(EPS_t)$$

Standardized unexpected earnings

$$SUE_t = \frac{EPS_t - E(EPS_t)}{\sigma[EPS_t - E(EPS_t)]}$$

LOS: Explain harmonic, weighted harmonic mean

Harmonic mean

$$X_H = \frac{n}{\sum_{i=1}^n (1/X_i)}$$

Weighted harmonic mean

$$X_{WH} = \frac{1}{\sum_{i=1}^n (w_i/X_i)}$$

LM5: Residual Income Valuation

LOS: Calculate/Interpret RI, EVA, MVA

Residual Income = NI - estimated cost of equity capital

$$EVA = NOPAT - (C\% \times TC)$$



MVA = Market value of company - Accounting book value of total capital

NOPAT	Net Operating Profit after Taxes
C	Cost of Capital (%)
TC	Total Capital

LOS: Calculate IV of a stock using RI models

General RI model:
$$V_0 = B_0 + \sum_{t=1}^{\infty} \frac{E_t - rB_{t-1}}{(1+r)^t}$$

or/
$$V_0 = B_0 + \sum_{t=1}^{\infty} \frac{(ROE_t - r)B_{t-1}}{(1+r)^t}$$

Single-stage (constant-growth) RI model

$$V_0 = B_0 + \frac{ROE - r}{r - g} B_0$$

Multi-stage RI model

$$V_0 = B_0 + \sum_{t=1}^T \frac{E_t - rB_{t-1}}{(1+r)^t} + \frac{P_T - B_T}{(1+r)^T}$$

or/

$$V_0 = B_0 + \sum_{t=1}^T \frac{(ROE_t - r)B_{t-1}}{(1+r)^t} + \frac{P_T - B_T}{(1+r)^T}$$

with persistence:

$$V_0 = B_0 + \sum_{t=1}^{T-1} \frac{E_t - rB_{t-1}}{(1+r)^t} + \frac{E_T - rB_{T-1}}{(1+r-\omega)(1+r)^{T-1}}$$

V₀	Value of a share of stock today (t = 0)
B₀	Current per-share book value of equity
B_t	Expected per-share book value of equity at any time t $B_t = B_{t-1} + E_t - D_t$
r	Required rate of return on equity investment (cost of equity)
E_t	Expected EPS for period t
ω	Persistence factor (between 0 and 1)

- A persistence factor of 1 implies that residual income will not fade at all; rather it will continue at the same level indefinitely. A persistence factor of 0 implies that residual income will not continue after the initial forecast horizon.

LM6: Private Company Valuation

LOS: Compare models used to estimate the required rate of return

Discount rates:

Expanded CAPM: $r_e = R_f + \beta ERP + \text{size premium} + \text{risk premium} + \text{company-specific risk premium}$

Build-up Approach: $r_e = R_f + ERP +/- \text{one or more premia (discounts)}$



(risk premia examples; size, industry, company-specific, etc..)

LOS: Calculate the value of a private company

Excess Earnings Method

$$EE = \text{Earnings} - r_{WC}(WC) - r_{FA}(FA)$$

$$V_{intg} = \frac{EE(1+g)}{r_{intg}-g}$$

Usually only used on smaller companies

- r_{WC} , r_{FA} and r_{intg} not realistic

$$V_f = WC + FA + \text{Intangible Assets}$$

EE	Excess earnings
r_{WC}	Cost of working capital
WC	Working capital
r_{FA}	Cost of fixed assets
FA	Fixed assets
V_{intg}	Value of intangible assets
r_{intg}	Cost of intangible assets
V_f	Value of the firm

Key multiple: MVIC/EBITDA

where: $MVIC = MV_e + MV_d$

MVIC	Market value of invested capital
MV_e	Market value of equity
MV_d	Market value of debt

Guideline Public Company or Transactions Method

(Guideline Public Company MVIC/EBITDA) x Normalized EBITDA – Debt Capital

Discount for Lack of Control

$$DLOC = 1 - [1/(1 + \text{Control Premium})]$$

Fixed Income

LM1: The Term Structure and Interest Rate Dynamics

Discount Factor: $DF_N = \frac{1}{(1 + z_N)^N}$

Forward Pricing Model: $DF_B = DF_A \times F_{A,B-A}$

Forward Rate Model: $(1 + z_B)^B = (1 + z_A)^A (1 + f_{A,B-A})^{B-A}$

$$\text{Swap Spread}(T) = \text{Risky Yield}(T) - \text{Default-Free Yield}(T)$$



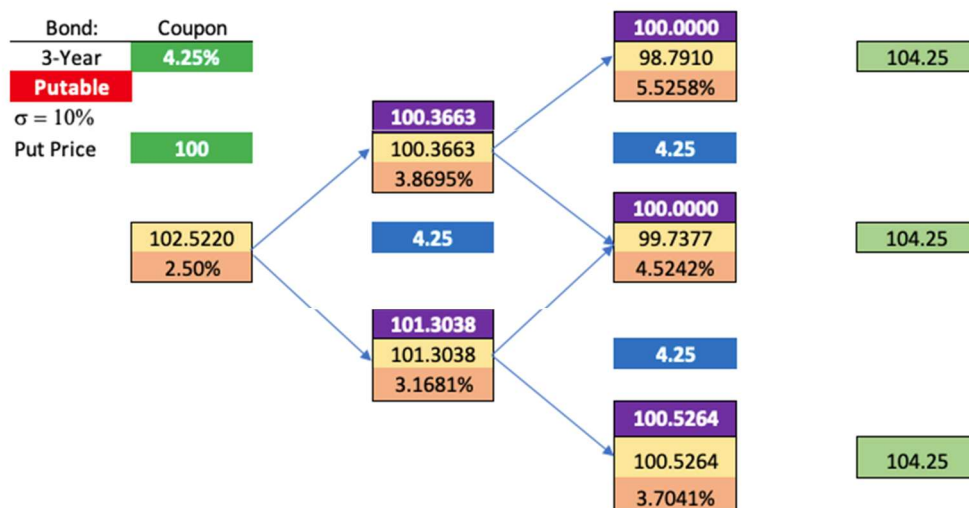
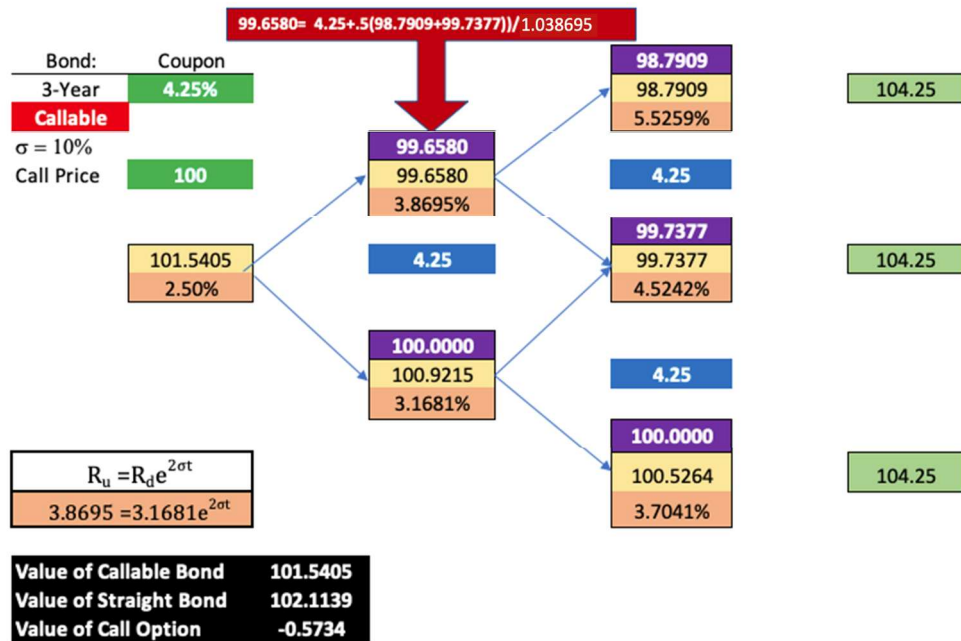
Risky Yield
Default-Free Yield

Typically a Corporate Bond/Note
Typically Treasury Yields

- The Treasury yield is arrived at by simple interpolation. The swap spread is the Corporate bond yield less the interpolated Treasury yield.
- SWAPS+ refers to the interpolated Treasury yield + the swap spread

LM3: Valuation and Analysis: Bonds with Embedded Options

Calculate - callable and putable bonds





Value of Putable Bond	102.5220
Value of Straight Bond	102.1139
Value of Put Option	0.4081

Maturity value of the bond

Value of the bond

Discount Rate

Value of the bond to be discounted back.		
Coupon		
Par	Spot	Forward
2.5000%	2.5000%	2.5000%
3.0000%	3.0080%	3.5180%
3.5000%	3.5240%	4.5640%

R_u Upper rate

R_d Lower rate

σ Volatility

t Time

LOS: Calculate – Effective Duration

$$\text{Effective Duration} = \frac{(PV_-) - (PV_+)}{2 \times (\Delta \text{Curve}) \times PV_0}$$

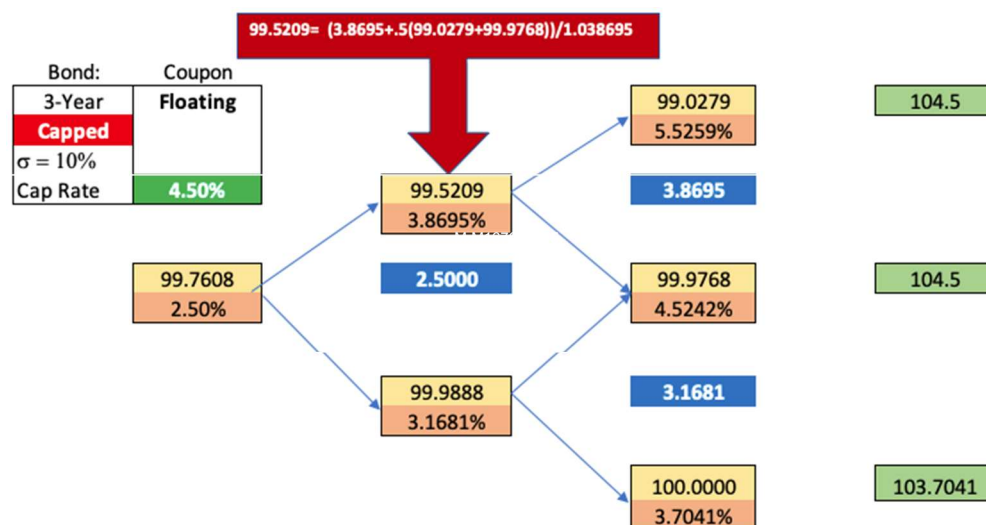
PV₋ the full price of the bond when the benchmark yield curve is shifted down by ΔCurve

PV₊ the full price of the bond when the benchmark yield curve is shifted up by ΔCurve

ΔCurve the magnitude of the parallel shift in the benchmark yield curve (in decimal)

PV₀ the current full price of the bond

LOS: Calculate - capped and floored floating rate bonds





Value of Capped Floater	99.7608
Value of Straight Bond	100.0000
Value of Embedded Cap	-0.2392

- The cap provision in a floater prevents the coupon rate from increasing above a specified maximum rate. As a consequence, a capped floater protects the issuer against rising interest rates and is thus an issuer option.

LOS: Calculate - components of a convertible bond's value

Conversion Ratio = Bond's Par Value/Conversion Price

Conversion Value = Share Price * Conversion Ratio

Minimum Value of a Convertible Bond = max(Conversion Value, PV of underlying straight bond)

Market Conversion Price = Convertible Bond price/CR

Market Conversion Premium per Share = Market Conversion Price - Underlying Share Price

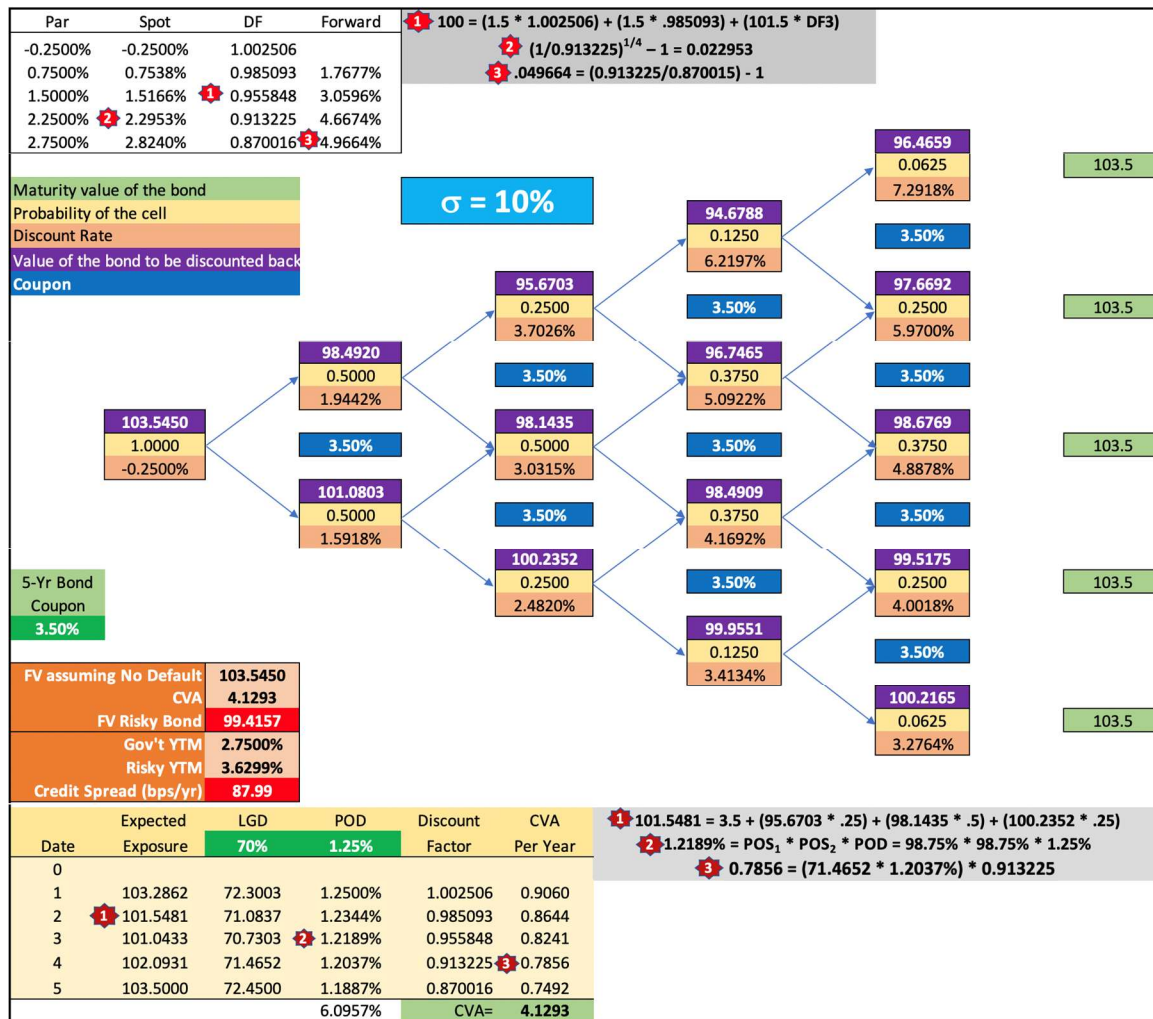
Market Conversion Premium Ratio = (MCP/sh)/Share Price

Premium over Straight Value = (Convertible Bond Price/Straight Value) – 1



LM4: Credit Analysis Models

Calculate - value of a bond, credit spread



LM5: Credit Default Swaps

LOS: Explain factors influencing pricing of CDS

Upfront payment = Present value of protection leg - Present value of premium leg

Upfront payment $\approx$ (Credit Spread – Fixed Coupon) x Duration

Profit for the buyer of protection $\approx$ Change in spread in bps x Duration x Notional



Derivatives

LM1: Pricing and Valuation of Forward Commitments

LOS: Calculate the no-arbitrage value of forwards

Value of Long Forward Contract Prior to Maturity

$$V_t = \frac{[F_t - F_0]}{(1 + r)^{T-t}}$$

Future Value of the underlying adjusted for carry cash flows

$$F_0 = FV[S_0 + CC_0 - CB_0]$$

Future value of the underlying with continuous compounding

$$F_0 = S_0 \exp^{(r_c + CC - C)T}$$

LOS: Calculate interest rate forwards and interpret their no-arbitrage value

FRA:

$$FRA_0 = \frac{\left\{ \left[\frac{1 + L_T t_T}{1 + L_h t_h} \right] - 1 \right\}}{t_m}$$

Receive floating

$$\frac{NA(\text{floating rate} - \text{fixed rate})(\text{days}/360)}{[1 + \text{floating rate}(\text{days}/360)]}$$

Receive fixed:

$$\frac{NA(\text{fixed rate} - \text{floating rate})(\text{days}/360)}{[1 + \text{floating rate}(\text{days}/360)]}$$

LOS: Calculate fixed-income forwards and interpret their no-arbitrage value

Fixed Income:

$$CTD = S_T - (Q_T \times CF)$$

$$F_0 = Q_0 \times CF$$

$$Q_0 = \frac{[(S_0 + AI) - I / (1+r)^T] (1+r)^T - AI}{CF}$$

CTD	Cheapest to Deliver
Q₀	Quoted Futures Price
CF	Conversion Factor



AI Accrued Interest

LOS: Calculate interest rate swaps and interpret their no-arbitrage value

Interest Rate Swaps:
$$PMT = \frac{1 - DF_n}{\sum DF}$$

Value of a swap: Receive fixed:
$$V_{\text{swap}} = (r_{\text{fix0}} - r_{\text{fixt}}) \sum DF \bullet NA$$

Value of a swap: pay fixed:
$$V_{\text{swap}} = (r_{\text{fixt}} - r_{\text{fix0}}) \sum DF \bullet NA$$

Currency Swap: value in domestic currency units

$$V_{\text{swap}} = V_D - S_{d/f} V_F$$

which leads to:
$$V_{\text{swap}} = NA(r_{\text{fxd}} \bullet \sum DF + DF_n) - S_{d/f}(r_{\text{fxf}} \bullet \sum DF + DF_n) NA$$

Currency Swap: value in foreign currency units

$$V_{\text{swap}} = V_D - V_D S_{f/d} - V_F$$

which leads to:
$$V_{\text{swap}} = NA(r_{\text{fxd}} \bullet \sum DF + DF_n) S_{f/d} - (r_{\text{fxf}} \bullet \sum DF + DF_n) NA$$

Equity swap, receive fixed

$$V_{\text{swap}} = V_{\text{fx}} - V_e$$

which leads to:
$$V_{\text{swap}} = NA(r_{\text{fx}} \bullet \sum DF + DF_n) - (S_t/S_0) NA$$

Equity swap, pay fixed

$$V_{\text{swap}} = V_e - V_{\text{fx}}$$

which leads to:
$$V_{\text{swap}} = NA(S_t/S_0) - NA(r_{\text{fx}} \bullet \sum DF + DF_n)$$

LOS: Calculate currency swaps and interpret their no-arbitrage value

Currency:

$$F_{P/B} = S_{P/B} \left(\frac{1 + i_P \left[\frac{\text{Actual}}{360} \right]}{1 + i_B \left[\frac{\text{Actual}}{360} \right]} \right)$$

with continuous rates:

$$F_{i/d} = S_{i/d} e^{(r_f - r_d)T}$$

LOS: Calculate equity swaps and interpret their no-arbitrage value

$$V_{EQ,t} = V_{FIX}(C_0) - \left(\frac{S_t}{S_{t-1}} \right) NA_E - PV(Par - NA_E)$$

LM2: Valuation of Contingent Claims

LOS: Calculate value of an option

Hedge ratio:

$$\Delta = \frac{c^+ - c^-}{s^+ - s^-} \quad \Delta = \frac{p^+ - p^-}{s^+ - s^-}$$



Risk neutral probability:

$$\pi_{up} = \frac{(1+r)-d}{u-d} \quad \text{and} \quad \pi_{down} = 1 - \pi_{up}$$

Value of an option: 1 period

$$c = \frac{\pi c^+ + (1-\pi)c^-}{(1+r)^T} \quad \text{AND} \quad p = \frac{\pi p^+ + (1-\pi)p^-}{(1+r)^T}$$

Value of an option: 2 periods

$$c = \frac{\pi^2 c^{++} + 2\pi(1-\pi)c^{+-} + (1-\pi)^2 c^{--}}{(1+r)^2} \quad \text{and} \quad p = \frac{\pi^2 p^{++} + 2\pi(1-\pi)p^{+-} + (1-\pi)^2 p^{--}}{(1+r)^2}$$

LOS: Describe how BSM values options

$$c = S \bullet N(d_1) - e^{-rT} X \bullet N(d_2)$$

$$p = e^{-rT} X \bullet N(-d_2) - S \bullet N(-d_1)$$

LOS: Describe how the Black model values options on futures

$$c = e^{-rT} [F_0(T) \bullet N(d_1) - X \bullet N(d_2)]$$

$$p = e^{-rT} [X \bullet N(-d_2) - F_0(T) \bullet N(-d_1)]$$

LOS: Describe how the Black model values interest rate options

$$c = NP \bullet [FRA_{T1,T2} \bullet N(d_1) - X_r \bullet N(d_2)] e^{-rT}$$

$$p = NP \bullet [X_r \bullet N(-d_2) - FRA_{T1,T2} \bullet N(-d_1)] e^{-rT}$$

cap = series of caplets (call options)

floor = series of floorlets (put options)

Long a floor, short a cap = receive fixed, pay floating swap

Swaptions

$$\text{Pay}_{\text{SWN}} = [R_{\text{fix}} \bullet N(d_1) - X_r \bullet N(d_2)] PV_A$$

$$\text{Rec}_{\text{SWN}} = [X_r \bullet N(-d_2) - R_{\text{fix}} \bullet N(-d_1)] PV_A$$

LOS: Describe how a delta hedge is executed

Optimal number of hedging units:

$$N_H = - \frac{\text{Portfolio delta}}{\text{Delta}_H}$$



Alternative Investments

LM1: Introduction to Commodities and Commodity Derivatives

Calculate components of total return

Price return = (current price - previous price)/previous price

$$\text{Roll return} = \frac{P_{\text{near}} - P_{\text{farther}}}{P_{\text{near}}} \times \%N_{\text{roll}}$$

Collateral return = HPY on cash/securities

Total Return = Price return + Roll return + collateral yield

P_{near}	Near-term futures contract closing price
P_{farther}	Farther-term futures contract closing price
%N_{roll}	Percentage of position in the futures contract being rolled

LM2: Overview of Types of Real Estate Investments

Direct Capitalization Method

Rental Income at full occupancy

+ Other income

= Potential Gross Income (PGI)

- Vacancy and collection loss

= Effective Gross Income (EGI)

- Operating expenses (OE)

→ (Excludes interest and taxes)

= Net Operating Income (NOI)

→ Forecasted first-year NOI for the property

Cap Rate = Discount Rate - growth rate

or/

Cap Rate = NOI/Sale price of comparable

Value = NOI/Cap Rate

- Note that the higher the growth rate, the lower the Cap Rate, the higher the value of the property. That is why the cap rate is called a *growth-implicit* capitalization rate. When applied to the NOI based on the first year of ownership, the cap rate is referred to as the going-in caprate.

Discounted Cash Flow (DCF) Method

Value = PV(Explicit Discount Period) + PV(Terminal Value)

Terminal Value = NOI/Terminal Cap rate



NOI
Terminal Cap Rate

The first year's NOI after the property is expected to be sold
Typically higher than the going-in cap rate (older properties may not be as competitive)

$$\text{DSCR} = \text{NOI} / \text{DS}$$

$$\text{LTV} = \text{Loan Amount} / \text{Appraised Value of property}$$

$$\text{Equity Dividend Rate} = \text{cash flow} / \text{Equity}$$

DSCR	Debt Service Coverage ratio, should be 1.2 or higher
DS	Debt Service (principal + interest payment)
LTV	Loan-to-Value Ratio
Cash Flow	NOI - Debt Service
Equity	Price of property - mortgage balance

Leveraged IRR:

PV = -Equity
PMT = cash flow
FV = sales price - mortgage balance
N = holding years
CPT I/Y

Unleveraged IRR:

PV = -Price
PMT = NOI
FV = sales price
N = holding years
CPT I/Y

LM3: Investments in Real Estate Through Publicly Traded Securities

LOS: Net asset value approach, P/FFO, and P/AFFO

Net Asset Value Approach

Last 12-months real estate NOI

- Non-cash rents

+ Adjustments for full impact of acquisitions

= pro forma cash NOI for the last 12 months

+ next 12 months growth in NOI

= Estimated next 12 months cash NOI

Divided by assumed cap rate

= Estimated value of operating real estate

+ Cash/Cash-Equivalents

+ Land held for future development

+ Accounts Receivable

+ Prepaid/Other Assets

= Estimated gross asset value

- Total debt

- Other liabilities

= Net Asset Value

divided by shares outstanding

= NAVPS



- **Non-cash rents:** Difference between the average contractual rent over the leases' terms and the cash rent actually paid
- **Adjustments for full impact of acquisitions:** NOI is increased to reflect a full year's rent for properties acquired during the course of the year
- **Assumed cap rate:** Estimated based on recent transactions for comparable properties

Relative Value (Price Multiples) Approach

P/FFO

where **FFO** = Net Income + Depreciation + deferred taxes + losses – gains

FFO/sh = FFO/Share count

P/AFFO

where **AFFO** = FFO – non-cash rent adjustment – recurring maintenance-type CAPEX and leasing commissions

AFFO/sh = AFFO/Share count

LM4: Hedge Fund Strategies

Conditional Factor-Risk Model:

$$(R_{HF_i})_t = \alpha_i + \beta_i(ERP)_t + D_t\beta_i(ERP)_t$$

Can be expanded to any number of risk-factors

where:

R_{HF} = return to hedge fund i

D_t = dummy variable for time period t – the conditional

Portfolio Management

LM2: Using Multi-Factor Models

LOS: Carhart 4-Factor Model (as an example of a Fundamental Factor Model)

$$E(R_p) = R_F + \beta_1 RMRF + \beta_2 SMB + \beta_3 HML + \beta_4 WML$$

- Fundamental → risk factors are premiums

$$E(R_i) = a_i + b_{i1}F_{INFL} + b_{i2}F_{GDP} + \varepsilon_i$$

- Macroeconomic - risk factors are surprises (actual - expected)

E(R_p)	expected return
RMRF	market risk premium (market - risk-free rate)
SMB	small cap risk premium (small - big)
HML	value risk premium (high book-to-market – low book-to-market)
WML	momentum premium (winners - losers)
b_i	factor sensitivities (regression slope coefficients)



R_f	risk-free rate
a_i	intercept, which is the expected return on asset i
ε_i	error term, represents the portion of R_i not explained by the factor model
F_{INFL}	inflation surprise (actual inflation - expected inflation)
F_{GDP}	GDP growth surprise (actual GDP growth - expected GDP growth)

notes:

- The Carhart model can be viewed as a multifactor extension of the CAPM that explicitly incorporates drivers of differences in expected returns among asset variables that are viewed as anomalies from a pure CAPM perspective. It is unlikely that all coefficients would have the same sign since value stocks are rarely momentum stocks and small-cap stocks are rarely value stocks. Large cap stocks will tend to have a beta coefficient of 1 or less for RMRF but a coefficient greater than 1 for small cap stocks.
- The INFL risk factor value is typically negative in a time series, so a positive coefficient would lead to a lower expected return and a negative coefficient would lead to a higher expected return.

LM3: Measuring and Managing Market Risk

LOS: Value at Risk (VaR)

$$\text{Annual VaR} = [E(R_p) - z\sigma_p](-1)NA$$

$E(R_p)$	Expected return of the portfolio
σ_p	Standard deviation (volatility) of the portfolio
z	z-value associated with the threshold VaR value
NA	Notional amount

- The expected return of the portfolio on a daily $\frac{E(R_p)}{250}$, weekly $\frac{E(R_p)}{52}$, monthly $\frac{E(R_p)}{12}$ basis.
- The standard deviation (volatility) of the portfolio on a daily $\frac{\sigma_p}{\sqrt{250}}$, weekly $\frac{\sigma_p}{\sqrt{52}}$, monthly $\frac{\sigma_p}{\sqrt{12}}$ basis.
- z-value associated with the threshold VaR value; 1% = 2.33, 5% = 1.65, 1s = 16% = 1

LM5: Economics and Investment Markets

Describe only

$$P_t^i = \sum_{s=1}^N \frac{E_t[CF_{t+s}]}{[1+I_{t,s}+\Theta_{t,s}+\pi_{t,s}+\gamma_{t,s}^i+k_{t,s}^i+\phi_{t,s}^i]^s}$$

P	Present Value
$E[CF]$	Expected uncertain cash flows
I	Real default-free interest rate
Θ	Inflation expectations
γ	Default risk (credit spread)
κ	Equity risk premium
Φ	Liquidity risk premium
π	Uncertainty regarding inflation expectations



LM6: Analysis of Active Portfolio Management

LOS: Describe only

Measuring Active Return: $R_A = \sum_{i=1}^N \Delta w_i R_{Ai}$

Decomposing Active Return: $R_A = \sum_{j=1}^M \Delta w_j R_{B,j} + \sum_{j=1}^M w_{p,j} R_{A,j}$

Decomposing Active Return with stocks and bonds as an example

$$R_A = (\Delta w_{stocks} R_{B,stocks} + \Delta w_{bonds} R_{B,bonds}) + (w_{P,stocks} R_{A,stocks} + w_{P,bonds} R_{A,bonds})$$

(value added from asset allocation) (value added from security selection)

R_A	active return
Δw_j	active weight for asset class j ($w_{pj} - w_{bj}$)
$R_{B,j}$	benchmark return on asset class j
$w_{p,j}$	portfolio weight on asset class j
$R_{A,j}$	$= R_j - R_{B,j}$

LOS: Sharpe ratio versus Information ratio

$$SR_p = (R_p - R_f) / \sigma_{Rp}$$

$$IR = R_A / \sigma_{RA}$$

Investing in the actively managed and benchmark portfolios to adjust Active Risk:

$$SR_p^2 = SR_B^2 + IR^2, \text{ and the optimal amount of active risk} = (IR / SR_B) \sigma_B \quad (\text{LOSd})$$

The portfolio with the highest IR will have the highest SR

SR_p	Sharpe ratio of the portfolio
R_p	return on the portfolio
R_f	risk-free rate
σ_{Rp}	standard deviation of the portfolio
IR	information ratio
R_A	active return
σ_{RA}	active risk (standard deviation of the active return)

- The Sharpe ratio is an absolute reward-to-risk measure. Ex-post uses R_p and ex-ante uses $E(R_p)$. Both the return and the standard deviation are annualized in the calculation. The Sharpe Ratio is unaffected by the addition of cash or leverage in a portfolio.
- An assumption is made with this formulation that the beta of the managed portfolio with respect to the benchmark = 1. Both active return and active risk are annualized. Ex-post uses R_A and ex-ante uses $E(R_A)$. The IR is affected by the addition of cash or leverage to the portfolio but is unaffected by the aggressiveness of the active weights.



LOS: Fundamental Law of Active Management

$$E(R_A) = (TC)(IC)\sqrt{BR}\sigma_A$$

dividing by σ_A

$$IR = (TC)(IC)\sqrt{BR}$$

E(R_A)	Expected return on an actively managed portfolio
TC	Transfer Coefficient
IC	Information Coefficient
BR	Breadth
σ_A	Active Risk

- TC = risk-weighted correlation between the forecasted active security returns and the actual active weights. An unconstrained portfolio will have TC = 1. BR = the number of independent decisions made each year. IC = the risk-weighted correlation between the forecasted actual returns and the realized active returns.